Business Requirements Document (BRD), Version 2.0

Amrit Water WhatsApp AI Sales & Order Agent

Document	BRD v2.0
Project	Amrit Water WhatsApp AI Agent (upgrade of the v1 assistant)
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Status	Proof of concept. Demo rates and test data. Prepared for review; not signed off by a client.
Replaces	BRD v1 (fixed-flow assistant). v1 documents stay in the repository for reference.

1. Purpose and scope

This document defines the business requirements for version 2 of the Amrit Water WhatsApp solution: an AI agent that answers price questions, takes and confirms wholesale water orders, records them, and escalates sensitive cases to the owner. It covers ordering and enquiries on WhatsApp for a wholesale packaged-water business that sells 500 ml bottles in cartons of 12 within Ratlam, India.

2. Background and what v1 left open

Amrit Water receives orders and questions on WhatsApp. Version 1 turned this into a fixed-flow assistant: extract order details, apply business rules, save the order, and reply. Testing and review showed these gaps:

- General questions (price, delivery area) fell into an empty branch and received no reply.
- There was no memory between messages, so follow-ups such as 'make it 5' could not be handled.
- Incomplete orders created duplicate pending rows.
- The confirmation message was fixed text, and there was no way to hand a case to the owner.

3. Business objectives

#	Objective	How it is measured
O1	Give customers a fast, consistent first response	Replies within seconds (2 to 3.5 seconds observed in testing)
O2	Capture complete orders every time	All required fields collected before an order is saved; no duplicate rows
O3	Quote prices accurately	Zero invented prices or totals in the test set
O4	Keep the owner in control	Owner alerted for every confirmed order and escalation; confirmed orders are never cancelled by the bot
O5	Remove manual order logging	Every order stored with a status in a data table

4. Stakeholders

Stakeholder	Role	Interest
Business owner	Approves rules and rates; receives alerts	Accuracy, control, no wrong commitments to customers
Customers (retail and bulk buyers)	Order and ask questions in English, Hindi or Hinglish	Quick, clear answers and an easy way to order
Consultant	Designs, builds, tests and hands over	Delivery quality and a maintainable solution
Platform providers (Meta, n8n, model provider)	Messaging, orchestration and model services	Policy compliance and usage limits

5. Scope

In scope

- Order intake, price quotes, order confirmation and order status through WhatsApp
- Order and rate storage in data tables
- Owner alerts for confirmed orders, payment choice, complaints, discount requests, out-of-area delivery and cancellation requests
- English and Hinglish replies

Out of scope (for this version)

- Online payment collection, stock checks and delivery scheduling
- Delivery tracking or any claim about delivery progress
- Multiple products, multiple service areas and voice notes
- Automated refunds, discounts or custom pricing

6. Functional requirements

ID	Requirement	Priority	Test ref
FR1	Receive customer WhatsApp messages and reply on the same channel	Must	T1, T2
FR2	Collect order details one question at a time: bottle size, cartons, delivery location, customisation. The phone number is taken from the sender	Must	T2
FR3	Convert bottles to cartons only when divisible by 12; otherwise ask for whole cartons	Must	T6, T7
FR4	Quote prices only from the rate table by quantity slab	Must	T3, T8
FR5	Calculate every total with a calculator tool, including delivery	Must	T3, T6, T8
FR6	Save a draft order with status pending_confirmation once all fields are known	Must	T2, T3
FR7	Let the customer change the draft before confirming	Must	T3
FR8	Mark an order confirmed only after an explicit yes; ask for the payment method; notify the owner	Must	T4, T5
FR9	Cancel an unconfirmed draft on request	Should	T11
FR10	Never cancel a confirmed order automatically; send the owner a cancellation request and tell the customer	Must	T12
FR11	Report order status using only data in the orders table	Should	T13
FR12	Escalate complaints, discount requests and delivery outside Ratlam to the owner	Must	T9, T10

ID	Requirement	Priority	Test ref
FR13	Reply in the customer's language (English or Hinglish)	Should	T14
FR14	Remember the conversation separately for each customer	Should	T3, T14
FR15	Block malformed model output (for example leaked reasoning) before it reaches the customer	Must	Defect D7 regression

7. Business rules

ID	Rule
BR1	One carton contains 12 bottles. Only 500 ml is sold.
BR2	Delivery is within Ratlam only. Any other location is referred to the owner.
BR3	Delivery charge: ₹50 under 10 cartons; free from 10 cartons.
BR4	Demo rate card per carton: 1 or more cartons ₹108; 10 or more ₹102; 50 or more ₹96.
BR5	Prices are before GST.
BR6	An order is confirmed only after the customer explicitly says yes.
BR7	A confirmed order can be cancelled only by the owner.
BR8	The agent never states stock levels, delivery times or delivery progress, because no such data exists.

The rate card is an assumed demo rate based on typical small-manufacturer listings (about ₹8 to ₹10 per bottle). It is not a Ratlam quotation.

8. Non-functional requirements

Area	Requirement
Performance	Reply within about five seconds under normal load (2 to 3.5 seconds observed in sampled executions)
Availability	Depends on the n8n plan, Meta and the model provider. The proof of concept carries no uptime commitment
Privacy	Store only the phone number and order details; keep tokens and phone numbers out of the repository and screenshots
Language	English and Hinglish (Roman script), matching the customer's latest message
Maintainability	Rates change by editing a table row, without editing the agent prompt
Auditability	Every order has a row with a status; n8n keeps execution history for troubleshooting

9. Data requirements

Table	Columns
Amrit_waters_orders	phone_number, bottle_size, quantity_cartons, delivery_location, customisation, status (pending_confirmation, confirmed, cancelled)
Amrit_waters_prices	min_cartons, price_per_carton

10. Assumptions and constraints

- Rates and orders are demo data; the project is a proof of concept.
- The owner and customer use different WhatsApp numbers in production.
- WhatsApp allows free-form messages to a number only within 24 hours of that number's last message; owner alerts rely on this (templates are needed for production).
- The proof of concept ran on free or trial tiers of the model provider and n8n.
- One product (500 ml) and one service area (Ratlam).

11. Risks

Risk	Impact	Mitigation
Model invents prices, fields or delivery claims	Wrong commitments to customers	Rate table, Calculator tool, explicit prompt rules, reply check, testing
Model rate limits or outage	No reply to the customer	Pacing, shorter memory, fallback model and paid tier for production
Conversation memory lost on restart	Customer must repeat details	Database-backed memory for production
Owner alert blocked by WhatsApp's 24-hour rule	Owner misses an alert	Approved message templates; keep the owner number active
Access token expiry	Bot stops sending messages	Permanent system-user token and a rotation date
Workflow change not published	Old behaviour stays live	Pre-release checklist in the runbook

12. Acceptance criteria

1. All 'Must' requirements pass in scenario testing (see the Test Report v2).
2. No invented price or total appears in any test conversation.
3. No order is confirmed without an explicit customer yes.
4. A confirmed order is never cancelled by the bot.
5. Every escalation reaches the owner's phone.
6. A defect log is maintained, and each defect has a recorded fix and retest.

13. Changes from v1

Area	v1 assistant	v2 agent
Control flow	Fixed pipeline of extract, branch, save, reply	Agent chooses tools within rules
General questions	No reply	Answered, or escalated to the owner
Memory	None	Per-customer memory
Prices	Not available	Rate table plus Calculator tool
Orders	Rows saved on every order message, with duplicates	One draft per order, edited in place, confirmed by the customer
Human in the loop	None	Owner alerts and escalation rules
Output safety	None	Reply check before sending